MA2101 - Linear Algebra via Matrices

- **Matrix operations:** Basic operations, Transpose and adjoint, Elementary row operations, RREF, Determinant, Computing matrix inverse.
- **Systems of linear equations:** Linear independence, Rank, Solvability, Gauss-Jordan elimination and Gauss elimination.
- **Matrix as a linear map:** subspace and span, Basis & dimension, Linear transformation, Coordinate vectors, Coordinate matrices, Change of basis matrix, Equivalence and similarity.
- **Orthogonality:** Inner products, Gram-Schmidt process, QR-factorization, Orthogonal projection, Best approximation, Least squares solutions.
- **Eigenvalues and Eigen vectors:** The characteristic polynomial, spectrum, Special types of matrices.
- **Canonical forms:** Schur triangularization, Annihilating polynomials, Diagonalizability, Jordan form, Singular value decomposition.

Course References:

1. (Text book) A. Singh **AS**, Introduction to Matrix Theory, Springer, 2021.
2. (Reference book) G. Strang, Introduction to Linear Algebra, 6th Ed., Wellesley-Cambridge Press, 2023

Section-wise mapping of syllabus to book (AS) sections

1. **Matrix operations:** Chapter 1: §1.2–§1.7
2. **Systems of linear equations:** Chapter 2: §2.1-§2.5
3. **Matrix as a linear map:** §3.1-§3.7
4. **Orthogonality:** Chapter 4: §4.1–§4.6
5. **Eigenvalues and Eigen vectors:** §5.2–§5.4
6. **Canonical forms:** §6.1–§6.5

Exam Dates

Exam	Date	Tentative meeting date
Quiz 1	13 February (Thu)	7 February (Friday)
Quiz 2	24 March (Mon)	18 March (Tue)
End sem	5 May (Mon)	28 April (Mon)

Note

- Commencement of classes: **Monday (16 Jan 2025)**. Classes are in A slot. Mon (8:00–8:50 AM), Tue (1:00–1:50 PM), Thu (11:00–11:50 AM), Fri (10:00–10:50 AM).
- Class number

Instructor	Room number
Arindama Singh (AS)	CRC202
Sounaka Mishra (SM)	CRC203
Priyanka Shukla (PS)	CRC204
K. Sumesh (KS)	CRC205

- Last Instructional Day / Compilation of attendance: **2 May 2025**.
- Total class $\approx$ 43. 13 classes (before quiz 1) + 16 classes (before quiz 2) + 14 classes (before endsem).
- Time-line: For **Quiz 1** Chapters 1–2; **Quiz 2** Chapters 3–4; **End-sem** Chapters 5–6;